

React Concepts and Explanations

Components

Components are reusable building blocks of a React application. They encapsulate UI, logic, and behavior, making applications modular and maintainable.

JSX

JSX is a syntax extension that allows developers to write HTML-like code inside JavaScript. It improves readability and simplifies UI creation.

Props

Props are inputs passed from parent components to child components. They help make components dynamic and reusable.

State

State stores data that can change over time within a component. Updating state automatically refreshes the UI.

Hooks

Hooks such as `useState` and `useEffect` allow functional components to manage state and side effects efficiently.

Virtual DOM

The Virtual DOM is a lightweight copy of the real DOM. React updates only changed elements, improving performance.

Event Handling

React handles user actions like clicks and form submissions through event handlers attached to JSX elements.

Conditional Rendering

Conditional rendering allows components to display different content based on conditions or application state.

React Router

React Router enables navigation between pages in a single-page application without reloading the browser.

Context API

Context API provides a way to share global data across components without excessive prop drilling.